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Sensitivity Analysis

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1 Introduction

Investment instruments play a crucial role in managing market risks and speculating future price movements. One of the most popular investment instruments is options, which gives the holder the right to buy or sell an underlying asset at a specific price and time. Traders use "Greeks," such as delta, gamma, and vega, to evaluate the risk and value of their option positions.

This report explores the Greeks of European vanilla options under the Black-Scholes model and a "pseudo-CEV" dynamic. Four numerical methods are utilized to compute the Greeks : a constant-step finite difference approach, a decreasing-step finite difference approach, a tangent process approach, and a Malliavin calculus approach.

The report begins by providing an overview of the Greeks and their definitions, followed by a theoretical analysis of each numerical method. We then present our implementation choices and analyze the sensitivities of each payoff under both the pseudo-CEV dynamic and a Geometric Brownian Motion.

Our study contributes to the existing literature on option pricing by offering a comprehensive comparison of different numerical methods for calculating sensitivities under multiple models. We also emphasize the significance of examining the sensitivity of options under different models and provide insights into the impact of model selection on option pricing.

2 Sensitivities (Greeks)

In the entire report, we will be studying financial derivatives and generally consider $f(\cdot)$ there corresponding payoff, as we have seen in the course we have there corresponding price using Feynman-Kac :

$$F(t, x, \sigma) = \mathbb{E}^{\mathbb{Q}}[e^{-r(T-t)} f(S_{T-t}^x)], \quad \forall t \leq T. \quad (1)$$

This price can be easily evaluated by a Monte Carlo method since it is easy to simulate S_{T-t}^x samples by simulating independent normal random variables. In addition, two different models are studied with regard to the dynamics of the underlying asset, in a first step the well-known Black & Scholes model :

$$\frac{dS_t^x}{S_t^x} = rdt + \sigma dW_t, \text{ with } S_0^x = x, \quad (2)$$

and in a second step, the « pseudo-CEV » model :

$$\frac{dS_t^x}{S_t^x} = rdt + \frac{(S_t^x)^\vartheta}{\sqrt{1 + (S_t^x)^2}} dW_t, \text{ with } S_0^x = x \text{ and } \vartheta \in]0, 1[. \quad (3)$$

The Euler scheme of the « pseudo-CEV » model is given by :

$$S_{t_i}^x = S_{t_{i-1}}^x + r S_{t_{i-1}}^x h + \frac{(S_{t_{i-1}}^x)^\vartheta}{\sqrt{1 + (S_{t_{i-1}}^x)^2}} \sqrt{h} N, \text{ with } h = (t_i - t_{i-1}) \text{ and } N \sim \mathcal{N}(0, 1) \quad (4)$$

Let's recall the formulas for delta, vega and gamma in the following table :

Payoff	Δ_t^{BS}	Γ_t^{BS}	v_t^{BS}
Call Option	$\mathcal{N}(d_+)$	$\frac{1}{S_t^x \sigma \sqrt{T-t}} \mathcal{N}'(d_+)$	$S_t^x \sqrt{T-t} \mathcal{N}'(d_+)$
Digital Put Option	$-\frac{e^{-r(T-t)}}{S_t^x \sigma \sqrt{T-t}} \mathcal{N}'(d_-)$	$\frac{e^{-r(T-t)}}{(S_t^x)^2 \sigma^2 (T-t)} \mathcal{N}'(d_-) d_+$	$\frac{\partial \Psi_t^{\text{BS}}}{\partial \sigma} = \frac{e^{-r(T-t)}}{\sigma} \mathcal{N}'(d_-) d_+$

TABLE 1 – Closed formulas of the Greeks

where $\mathcal{N}(x)$ is the Normal cumulative distribution function,

$$\mathcal{N}'(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}, \quad (5)$$

and,

$$d_{\pm} = \frac{\log\left(\frac{S_t^x e^{r(T-t)}}{K}\right) \pm \frac{1}{2} \sigma^2 (T-t)}{\sigma \sqrt{T-t}}. \quad (6)$$

For the pseudo-CEV model and as there exists no closed formulas for this case, we need to consider a « disturbed » process in order to compute the v . The v quantifies the impact, in a given direction, of a small disturbance in volatility on the price of an option, see [6] for more details about the subject. In other words, it is a derivative of the option price with respect to volatility in the sense of Gateaux.

Then, we need to introduce a « disturbed » process. Let $\tilde{\sigma} : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ the direction of the perturbation. Then, under the conditions that $\sigma(t, x)$ and $(\sigma + \epsilon \tilde{\sigma})(t, x)$ are continuously derivable with bounded derivatives, satisfying Lipschitz conditions and a uniform ellipticity relation, such that

$$\exists \eta > 0, |(\sigma + \epsilon \tilde{\sigma})(t, x)| \geq \eta, \forall x \in \mathbb{R}. \quad (7)$$

In order to respect all these conditions, we will use : $\tilde{\sigma} = S$, which seems to be a common choice in the literature. We can then introduce the « disturbed » (general) process :

$$dS_t^{x,\epsilon} = b(t, S_t^{x,\epsilon})dt + (\sigma + \epsilon \tilde{\sigma})(t, S_t^{x,\epsilon})dW_t, \quad (8)$$

where $S_0^{x,\epsilon} = x$.

The upcoming sections will examine various numerical methods. The first objective is to compare them with the exact value in the context of the Black-Scholes model, [1]. The second objective is to use these methods to derive a value that is as precise as possible for the pseudo-CEV model.

3 Technical Overview of Algorithmic Techniques

3.1 Finite Difference Methods for sensitivities calculation

For the classical Monte Carlo method, the calculation of the Greeks is done by a finite difference scheme. Thus, the value of Δ can be approximated by a slight variation in the starting condition. Three different schemes are possible : forward difference $\frac{F(t, x+h, \sigma) - F(t, x, \sigma)}{h}$, central difference $\frac{F(t, x+h, \sigma) - F(t, x-h, \sigma)}{2h}$ and backward difference $\frac{F(t, x, \sigma) - F(t, x-h, \sigma)}{h}$. We obtain the following expressions for the different Greeks :

$$\Delta_t(x) \approx \frac{F(t, x+h, \sigma) - F(t, x-h, \sigma)}{2h} \quad (9)$$

$$\Gamma_t(x) \approx \frac{F(t, x+h, \sigma) + F(t, x-h, \sigma) - 2F(t, x, \sigma)}{h^2} \quad (10)$$

$$v_t(x) \approx \frac{F(t, x, \sigma+h) - F(t, x, \sigma-h)}{2h} \quad (11)$$

Note that for the pseudo CEV, the vega is compute based on the « disturbed » process as mentioned previously. So if we consider $F(t, x, \varepsilon) = \mathbb{E}^{\mathbb{Q}}[e^{-r(T-t)} f(S_{T-t}^{x,\epsilon})] \big|_{\tilde{\sigma} \text{ given}}$ and then we have :

$$v_t^{CEV}(x) \approx \frac{F(t, x, \varepsilon = h) - F(t, x, \varepsilon = -h)}{2h} \quad (12)$$

Although the finite difference method is commonly used in numerical approximation, it has several limitations.

Firstly, it relies on a Monte Carlo method to obtain the price, which can introduce errors. Additionally, the approach of a numerical derivative can also cause inaccuracies. The combination of these two approximations often results in an increase in the overall error.

Furthermore, the precision of the finite difference method is highly dependent on the choice of the parameter h . If h is too large, the results can be significantly biased. Conversely, if h is too small, the variance of the estimator can explode.

To address these challenges, we will explore two potential approaches in the following sections. In the first subsection, we will examine the use of a constant pitch. In the second subsection, we will consider decreasing the pitch as the simulations progress to potentially improve accuracy.

To facilitate the interpretation of our results, we will use the notation $f(x, Y)$ to represent $f(S^x)$ and $F(x)$ to denote $F(t, x, \sigma)$. By adopting this notation, we aim to enhance clarity and simplify our analysis.

3.1.1 Fixed Step Method for sensitivities calculation

When using a constant step approach, Pagès (2018) demonstrates that if the payoff function f satisfies a local mean quadratic Lipschitz continuity assumption at x , where $C(f, Y)$ is a constant determined by the payoff and the random variable, certain properties can be derived.

Furthermore, assuming that F is twice differentiable with a Lipschitz continuous second derivative on the interval $(x - h_0, x + h_0)$, and considering a sequence of i.i.d. random vectors $(Y_k)_{k \geq 1}$ with the same distribution as Y , it can be shown that for any h within the range of $(0, h_0)$ we obtain the following expression :

$$\begin{aligned} \left\| F'(x) - \frac{1}{M} \sum_{k=1}^M \frac{f(x+h, Y_k) - f(x-h, Y_k)}{2h} \right\|_2 &\leq \sqrt{\left([F'']_{\text{Lip}} \frac{h^2}{2} \right)^2 + \frac{C^2(f, Y)}{M}} \\ &\leq \left([F'']_{\text{Lip}} \frac{h^2}{2} \right) + \frac{C(f, Y)}{\sqrt{M}}. \end{aligned} \quad (13)$$

Thus, it can be seen that the number of simulations and the step size are somehow linked and that it is not efficient to choose them arbitrarily, since $M := M(h) = \mathcal{O}(h^{-4})$. The effect of this relationship between these two parameters can be seen later in the results obtained.

3.1.2 Decreasing Step Method for sensitivities calculation

If a constant step is used, the sensitivity computation bias never goes away, and starting again with a new step size is necessary to enhance accuracy. To resolve this, one can employ a decreasing sequence of step sizes that gradually approaches zero as the number of simulations grows. Pagès (2018) suggested a method for doing this, which employs the same notation and process as the prior section.

$$\left\| F'(x) - \frac{1}{M} \sum_{k=1}^M \frac{f(x + h_k, Y_k) - f(x - h_k, Y_k)}{2h_k} \right\|_2 \leq \frac{1}{\sqrt{M}} \sqrt{\frac{[F'']_{\text{Lip}}^2}{4M} \left(\sum_{k=1}^M h_k^2 \right)^2 + C^2(f, Y)}. \quad (14)$$

Thus, in order to reach a $\sqrt{M} - L^2$ convergence as in standard classic Monte Carlo, we need to satisfied the following relationship :

$$\left(\sum_{k=1}^M h_k^2 \right)^2 = \mathcal{O}(M), \quad (15)$$

leading to choose the step h_k such as $M := M(h) = \mathcal{O}(h^{-1/4})$ as $k \rightarrow \infty$. As we will see in the results, this change in the choice of step will have a big impact on convergence.

3.2 Malliavin Calculus

The Malliavin calculus [8] is a technique that can be applied to infinite-dimensional spaces and is widely used in financial mathematics to compute derivative sensitivities. It eliminates the need to calculate the derivative of the payoff by using an infinite-dimensional integration by parts formula. Instead, one can estimate the expectation of the option's payoff multiplied by a specific weight, known as the "Malliavin weight." This approach saves time in numerical calculations and allows for a general Monte Carlo algorithm to be developed for any option, regardless of its complexity, including digital options with discontinuous payoffs.

3.2.1 Greeks Implementation

Black and Scholes. The method of weights in the Black-Scholes model doesn't require Malliavin calculus to calculate the Greeks because the Gaussian density is regular. Using standard calculus and integration by parts with $f(\cdot)$ as the payoff function, we get an explicit formula for the price given by :

$$F(t, x, \sigma) = e^{-r(T-t)} \int_{-\infty}^{+\infty} f\left(xe^{(r-\frac{\sigma^2}{2})(T-t)+\sigma y\sqrt{T-t}}\right) \frac{1}{\sqrt{2\pi}} e^{-\frac{y^2}{2}} dy.$$

Setting $h(x, y) := f\left(xe^{(r-\frac{\sigma^2}{2})(T-t)+\sigma y\sqrt{T-t}}\right)$, we can simplify the calculation by observing that $\frac{\partial h}{\partial x}(x, y) = \frac{1}{x\sigma\sqrt{T-t}} \frac{\partial h}{\partial y}(x, y)$.

We can now derive the delta :

$$\Delta_t(x) = \frac{e^{-r(T-t)}}{x\sigma\sqrt{T-t}} \int_{-\infty}^{+\infty} f\left(xe^{(r-\frac{\sigma^2}{2})(T-t)+\sigma y\sqrt{T-t}}\right) \frac{y}{\sqrt{2\pi}} e^{-\frac{y^2}{2}} dy.$$

Finally, using the fact that $W_{T-t} \stackrel{d}{=} \sqrt{T-t} \times Z$ where $Z \sim \mathcal{N}(0, 1)$, we can express the delta as :

$$\Delta_t(x) = \frac{e^{-r(T-t)}}{x\sigma(T-t)} \mathbb{E}^{\mathbb{Q}} \left[W_{T-t} f(S_{T-t}^x) \right].$$

To calculate Γ using Monte Carlo, we can use the property that relates the second-order partial derivatives of the option payoff function with respect to x and y . By applying this property, we obtain a closed expression for Γ in terms of the expectation of the option payoff function and the moments of the standard normal distribution. Similarly, we can obtain a closed expression for v in terms of Γ and the moments of the standard normal distribution. This method does not require numerical derivatives and is not dependent on the option payoff function, only on the derivative. For more complex models, such as the "pseudo-CEV" model, we need to use integration by parts formulas based on the duality between the Malliavin derivative and the Skorokhod integral.

$$\Gamma_t(x) = \frac{e^{-r(T-t)}}{x^2\sigma(T-t)} \mathbb{E}^{\mathbb{Q}} \left[\left(\frac{W_{T-t}^2}{\sigma(T-t)} - \frac{1}{\sigma} - W_{T-t} \right) f(S_{T-t}^x) \right] \quad (16)$$

and,

$$v_t(x) = e^{-r(T-t)} \mathbb{E}^{\mathbb{Q}} \left[\left(\frac{W_{T-t}^2}{\sigma(T-t)} - \frac{1}{\sigma} - W_{T-t} \right) f(S_{T-t}^x) \right]. \quad (17)$$

The Monte Carlo algorithm can be used to approximate the derivative without relying on numerical derivatives, unlike the finite difference technique. Additionally, the method can be applied to any option payoff by changing the function according to the derivative. For more complex dynamics, integration by parts formulas induced by the duality between the derivation operator in the Malliavin sense and the Skorokhod integral are required.

Pseudo-CEV. This paragraph is based on Fournié et al.'s [4] results on sensitivities, but we will use simplified expressions since we are not studying path dependent products. The sensitivities are represented as expectations with weights π that do not depend on the payoff. We derive a simplified expression for Delta by using the integration theorem and the process of first variations. We then seek a weight π that can fit into the dynamics and allow us to use the integration formula by part. We obtain a condition for the weight π and solve it using the set $\mathbb{T}$ to define the permissible weights. We represent Gamma

assuming that the payoff is twice differentiable and has bounded derivatives. We set $G := e^{-r(T-t)}f(S_{T-t}^x)$ and derive an expression for Gamma.

$$\text{Greek} = \mathbb{E}^{\mathbb{Q}}[e^{-r(T-t)}f(S_{T-t}^x) \times \pi], \quad (18)$$

And for the Δ :

$$\Delta_t(x) = \mathbb{E}^{\mathbb{Q}}[e^{-r(T-t)}f(S_{T-t}^x) \times \delta(w^{delta})] \quad (19)$$

for the Γ :

$$\begin{aligned} \Gamma &= \mathbb{E}^{\mathbb{Q}}[G(\delta(w^{delta})\delta(w^{delta}) + \delta(\partial_x w^{delta}))] \\ &= \mathbb{E}^{\mathbb{Q}}[G \times \delta(w^{delta}\delta(w^{delta}) + \partial_x w^{delta})] \end{aligned} \quad (20)$$

then

$$\Gamma = \mathbb{E}^{\mathbb{Q}}[G \times \delta(w^{gamma})] \quad (21)$$

3.3 Tangent process for sensitivities calculation

First of all, we consider the Δ . Recall that using local theorem¹(see appendix) we can interchanging differentiation and expectation such that if adding the payoff function and the coefficients of the SDE of S_t^x are regularity assumption, one can end up with the following formula :

$$\frac{\partial F(0, x, \sigma)}{\partial x} = \mathbb{E}\left(e^{-rT}f'(S_T^x) \frac{\partial S_T^x}{\partial x}\right) \quad (22)$$

Where $\frac{\partial S_t^x}{\partial x} := Y_t$ is called the tangent process. Let's take a general SDE :

$$dS_t^x = b(S_t^x)dt + \sigma(S_t^x)dW_t, \quad S_0^x = x \in \mathbb{R} \quad (23)$$

Then if $b(\cdot)$ and $\sigma(\cdot)$ are regular enough, we can define the tangent process by :

$$dY_t = Y_t(b'(X_t^x)dt + \sigma'(X_t^x)dW_t), \quad Y_0 = 1. \quad (24)$$

For the two dynamics considered we have :

1. With as function f the regular payoffs considered here i.e call/put

Black and Scholes : $Y_t = \frac{S_t^x}{x}$

Pseudo-CEV : $dY_t = rY_t dt + \frac{\sigma}{(1 + (S_t^x)^2)} \left(\vartheta (S_t^x)^\vartheta \sqrt{1 + (S_t^x)^2} - \frac{(S_t^x)^{\vartheta+1}}{\sqrt{1 + (S_t^x)^2}} \right) dW_t$ (25)

This technique is limited to regular payoffs, specifically call/put payoffs, and is not applicable to binary call/put options. The method is unsuitable for calculating the gamma of call/put options as the first derivative of the payoff lacks regularity. However, it can be utilized to calculate the vega of call/put options.

In a Black and Scholes framework, we can apply the same methodology as we did for Δ .

$$\frac{\partial F(0, x, \sigma)}{\partial \sigma} = \mathbb{E} \left(e^{-rT} f'(S_T^x) S_T^x (W_T - \sigma T) \right) \quad (26)$$

We set $\partial_\epsilon S_t^{x,\epsilon} = Z_t^{x,\epsilon}$, such as, $Z_t^{x,\epsilon}$ is variation process of the process with respect to ϵ and its dynamic is given by :

$$\begin{aligned} Z_0^\epsilon &= 0 \\ dZ_t^\epsilon &= b'(S_t^{x,\epsilon}) Z_t^\epsilon dt + \tilde{\sigma}(S_t^{x,\epsilon}) dW_t + (\sigma' + \epsilon \tilde{\sigma}')(S_t^{x,\epsilon}) Z_t^\epsilon dW_t \end{aligned} \quad (27)$$

The tangent process approach can be used to compute the vega of the pseudo-CEV for call and put payoffs using Monte Carlo simulations by using the last equations.

4 Variance reduction

This section describes the three variance reduction methods used in the study. These methods include antithetic variables, control variate, and a combination of both. The antithetic variable method is applied to all greeks, payoffs, and dynamics (Pseudo-CEV or BS). On the other hand, the control variate method is only used for the delta of call/put payoffs and is based on call/put parity. The antithetic variable and control variate method combination takes advantage of both methods. The objective of these methods is to compute the expectation m_X of a variable X using Monte Carlo with a lower variance than $\mathbb{V}\text{ar}(X)$.

4.1 Control variate

The following introduces the control variate technique for estimating option Greeks using Monte Carlo simulations. The method involves two variables, X and X' , with the same expected value but different variances. A new variable, Ξ , is defined such

that $\mathbb{E}(\Xi) = 0$ and $\text{Var}(\Xi) > 0$. We calculate λ_{min} , which minimizes the variance of $X^\lambda = X - \lambda\Xi$, using λ and M simulations, and λ_{min} is estimated as $\hat{\lambda}M = \frac{C_M}{V_M}$. The approach is then used to compute Δ via call/put parity, with Ξ defined as $e^{-rT} \frac{dS_T}{ds_0} - 1$. The computation of Δ is summarized in two steps : (1) estimation of $\hat{\lambda}M'$ and (2) calculation of the Monte Carlo expectation. The true batch implementation is employed to estimate λ_{min} and reduce variance. The tangent process is utilized to simulate Ξ .

4.2 Antithetic variable

If two variables, X and X' , have the same mean and variance but are negatively correlated, the simulation of their average, Ξ , can decrease variance. The variance of Ξ is equal to the average of the variances of X and X' , plus their covariance. To achieve the same level of confidence, twice the number of simulations are required for Ξ compared to X . To calculate the Monte Carlo expectation of $X(W)$, $X(W^k)$ and $X(-W^k)$ are simulated for $k = 1, \dots, M$, and their average is computed. This technique of variance reduction can decrease variance by a factor of two.

4.3 Antithetic variable and Control variate

Combining Antithetic variable and Control variate can lead to higher variance reduction. As :

Compute $\hat{\lambda}_{M'}$ using the classical control variate method. Simulate W^k for $k = M', \dots, M$. Evaluate $X_k(W^k)$, $X_k(-W^k)$, $\Xi_k(W^k)$ and $\Xi_k(-W^k)$ for $k = M', \dots, M$. Compute the Monte Carlo expectation using the above evaluations and $\hat{\lambda}_{M'}$.

This method has been found to provide better variance reduction, see [9] for more details about the subject.

5 C++ Codes

We have decided to organize our code by dividing it into different HPP files, each defining a set of classes used in our program. In the *BS.hpp* and *Pseudo-CEV.hpp* files, we define the Black and Scholes dynamic class and the Pseudo-CEV dynamic class, respectively. These classes define the characteristics of the dynamic necessary for computing the greeks using various methods. Within each class, we have functions that generate simulations using Normal noise for the finite difference method, compute the tangent process, and provide Malliavin weights for Δ , Γ , and v . We use the same function names

for both classes to enable generic programming for the method classes.

We also define a separate class for each method in the *diff-finie.hpp*, *tangent-process.hpp*, and *malliavin.hpp* files, utilizing generic programming to allow support for both the Black and Scholes and Pseudo-CEV dynamic classes. These method classes have similar structures and contain three functions : *Delta*, *Gamma*, and *vega*. These functions generate payoffs from Normal noise inputs, which are then used to compute Monte Carlo expectations.

In the *MC.hpp* file, we use the *mean-var* class (defined by V.Lemaire) that returns empirical expectation, variance, and confidence interval length from a sum as input. Using this class, we can compute Monte Carlo estimators for the greeks.

In the *monte-carlo.hpp* file, we define four functions (*monte-carlo*, *monte-carlo-anti*, *monte-carlo-con-var*, and *monte-carlo-con-var-anti*) that utilize all the classes defined previously. These functions return a *mean-var* type object and execute the Monte Carlo algorithm to compute the greeks. We use generic programming to support the computation of different greeks and take as input a pointer to a function of a method class, allowing us to support functions Δ , Γ , and v from any of the methods. In these Monte Carlo functions, we generate Normal noise, which is used to obtain realizations of the random variables necessary for Monte Carlo estimators.

6 Numerical Results

6.1 Variance Reduction

The figure depicted in Figure 1 showcases the advantages of decreasing variance in the evaluation of greeks. The graph presents the computed value of a call option using the finite difference method with diverse variance reduction techniques. The outcomes demonstrate that implementing antithetic variables and control variates independently leads to a substantial reduction in variance. Furthermore, combining both methods leads to an even greater reduction in variance. The results presented in Table 2 validate that utilizing antithetic variables diminishes variance by two, while combining antithetic variables and control variates reduces the original variance.

The graph displayed in Figure 1, specifically the second graph, highlights the significant advantages of using the Call/Put parity relationship to minimize variance. The

Dynamic	No Variance reduction	Antithetic	Control variate	Antithetic + Control variate
BS ($M = 10^6$)	0.0015085	0.00069113	0.00034813	0.00034812
Pseudo-CEV ($M = 10^5$)	0.0053372	0.00301819	0.00192247	0.00102373

TABLE 2 – Confidence interval size of Monte Carlo estimator for Δ call with $h = 10^{-2}$

assessment of different methods is more effective for Put options than Call options due to the fact that in the case of Call options, the weights (i.e., the prices) can become extremely large since the value of the option is not upper-bounded. Conversely, Put options possess both upper and lower bounds (assuming no negative price), thus making their evaluation more efficient.

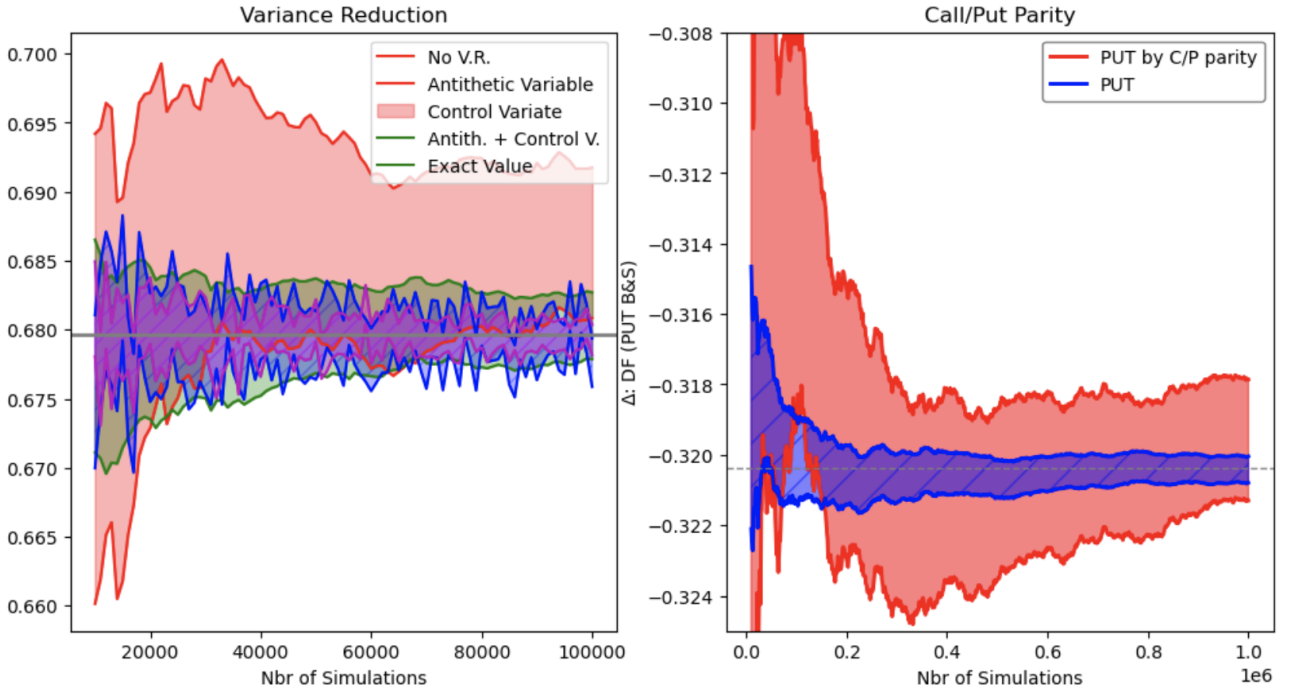


FIGURE 1 – Variance reduction study based on methods

6.2 Delta : Δ

To ensure consistency in the comparison of the graphs, it is important to note that the BS model was simulated up to 10^6 times, while the CEV model was only simulated up to 10^5 times. Additionally, all the results presented in the graphs are computed using the following parameters : $K = 100, S_0^x = 100, r = 10\%, \sigma = 60\%, T = 1, t = 0$ and for Pseudo-CEV $\vartheta = 0.9$.

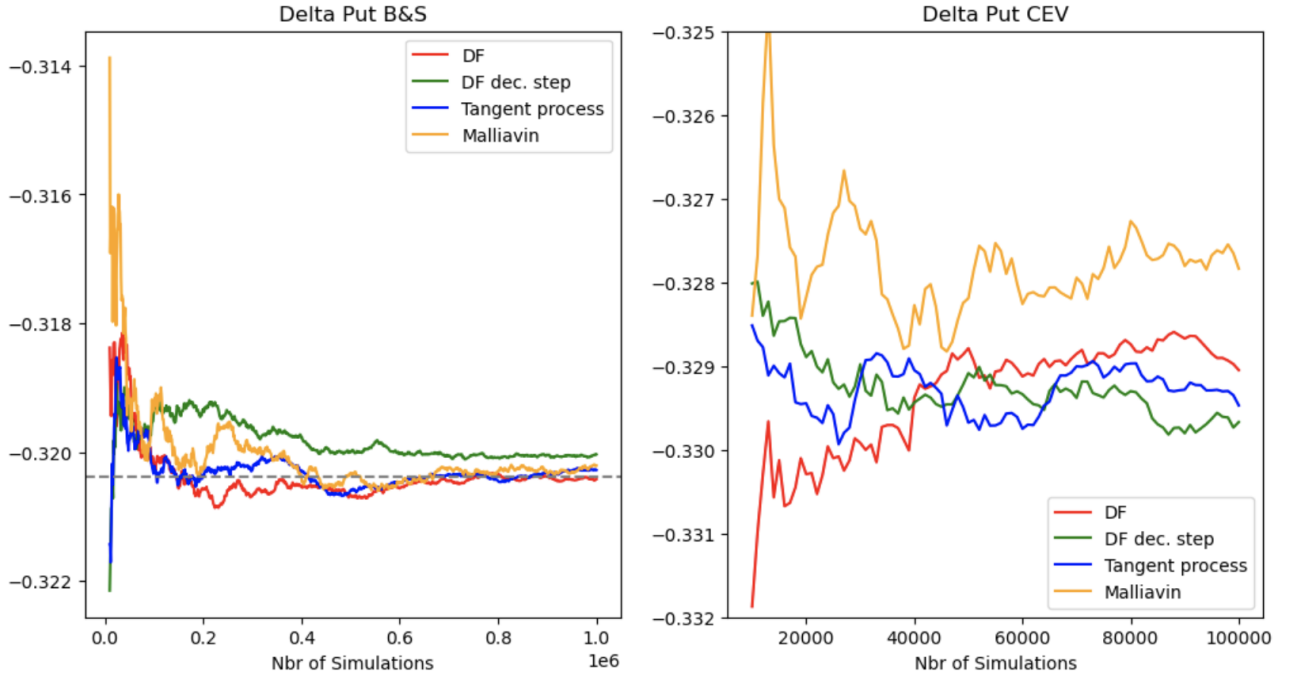


FIGURE 2 – Convergence of the different methods for the Put Delta

The study presented in Figure 2 examines various methods used for calculating the Δ Put. The findings indicate that the tangent flow approach yields promising outcomes, owing to its easy approximation in BS's case through the exact formula of the price dynamics. Conversely, the weight method does not offer superior precision compared to the finite difference method, as the Δ of a Put is a regular function. Consequently, the results for the CEV model are quite alike, making it challenging to draw definitive conclusions.

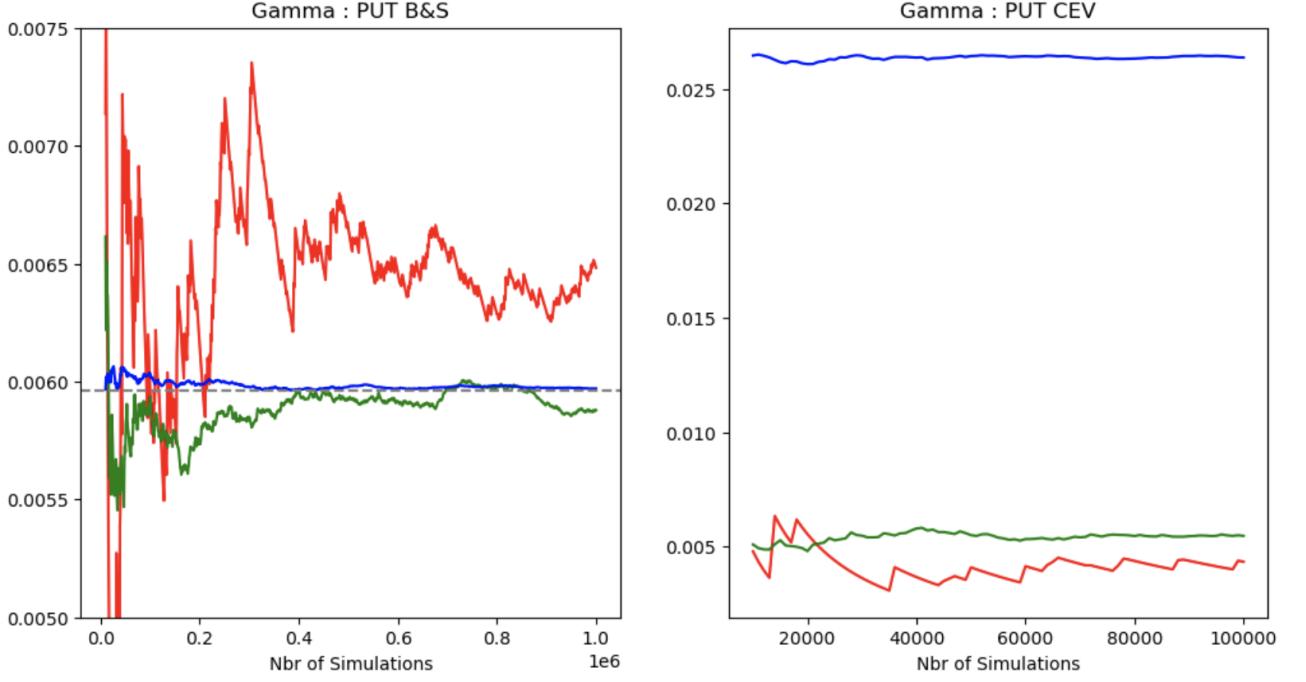


FIGURE 3 – Convergence of the different methods for the Digital Put Delta

Figure 3 presents the results obtained in the case of the binary put. We observe that when the payoff becomes discontinuous, the results are significantly better in the case of the Malliavin method, as previously announced in the report.

6.3 Gamma : Γ

The findings for the Γ Put are displayed in Figure 4. The graph shows that the weights method produces more precise results as the degree of differentiation increases, which is not surprising.

However, the use of a decreasing step also enhances the accuracy of the results and brings them closer to the actual value. The results for the CEV model are not as satisfactory for either method due to its higher complexity and fewer simulations. Nonetheless, the weights method still appears to be the better option.

The results for Gamma Binary Put are shown in Figure 4. The CEV model was very inaccurate using finite difference method, while the weights method and the decreasing step finite difference method were comparable. For the Black and Scholes model, both the weights method and decreasing step finite difference method converged to the exact value, but the Malliavin method was more accurate with fewer simulations needed for high accuracy on a smaller scale.

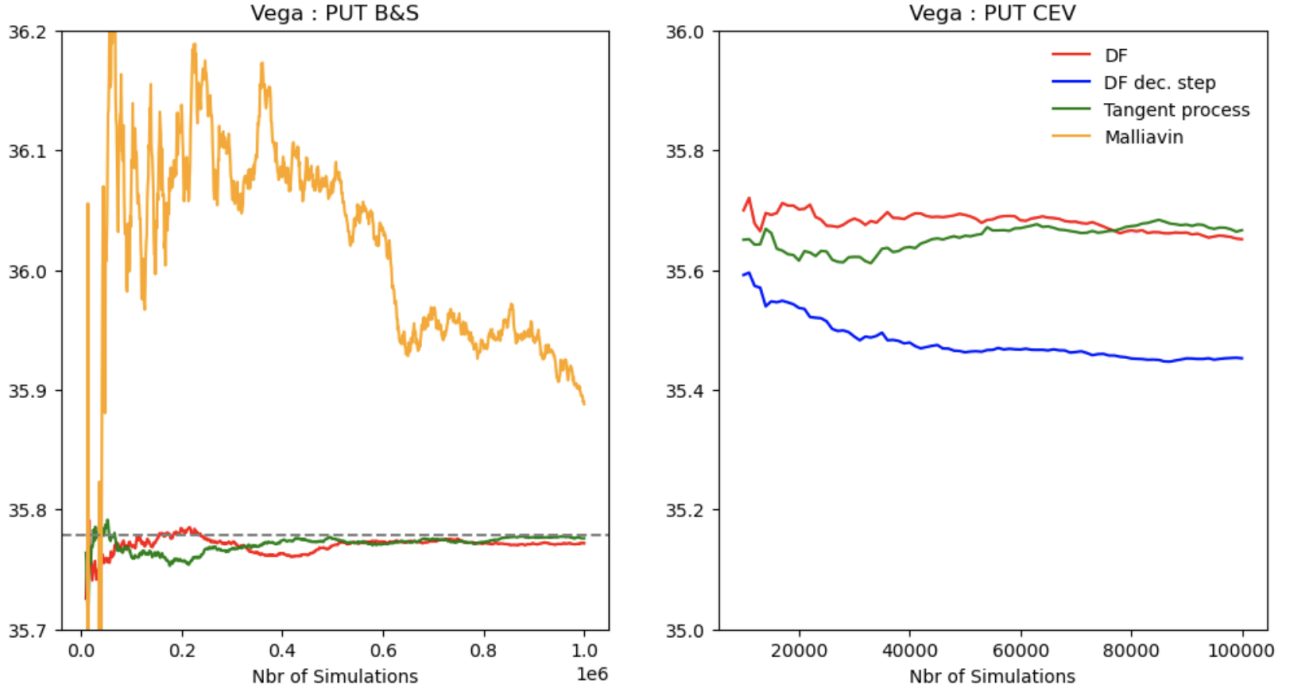


FIGURE 4 – Convergence of the different methods for Put Gamma

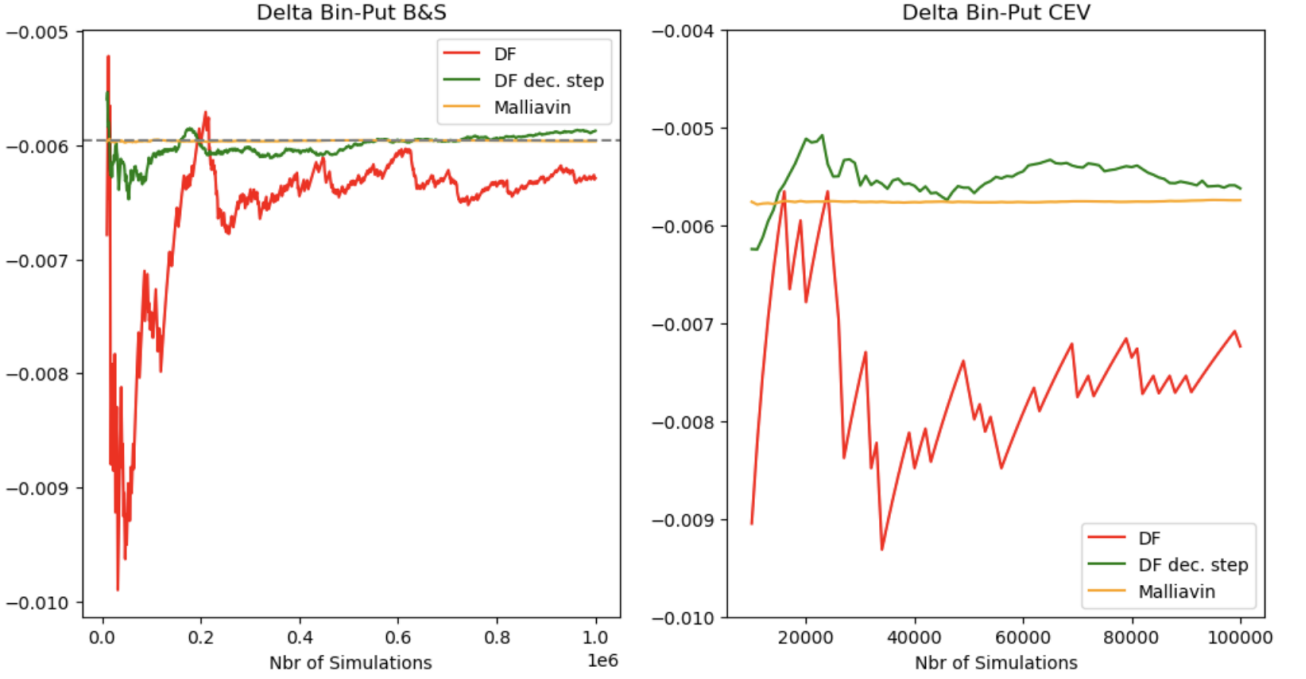


FIGURE 5 – Convergence of the different methods for Binary Put Gamma

6.4 Vega : v

Figure 6 displays the results of the v Put. Compared to other methods for this first-order Greek, the Malliavin approach is not as advantageous. However, the finite difference

method with decreasing step and the tangent flow method show promising outcomes. It is important to note that the scale is different and the values of v are higher, but the results are similar to those obtained for Δ .

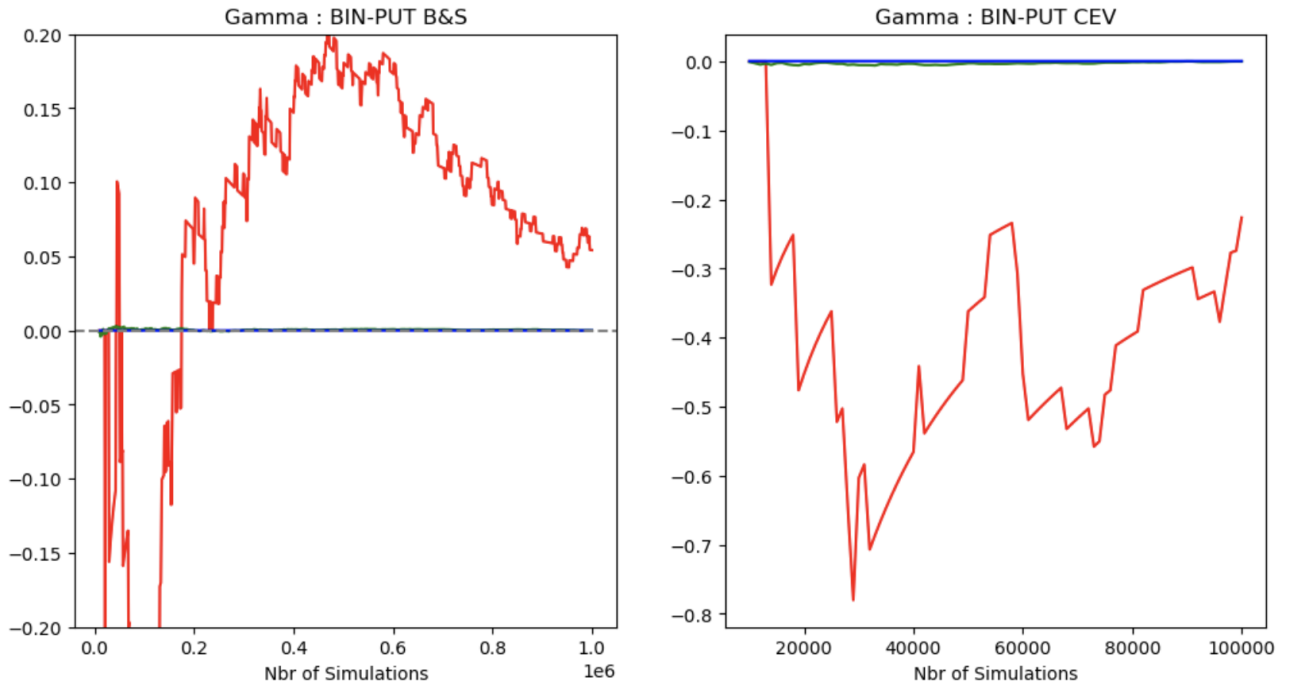


FIGURE 6 – Convergence of the different methods for Put Vega

Figure 7 shows results for v Binary Put. The weight method is once again the most accurate when dealing with a discontinuous payoff even for first order greeks. It is closest to the true value in the BS case and seems to be the most stable when used with the method of decreasing step finite differences in the CEV model, although the accuracy of the value obtained is uncertain.

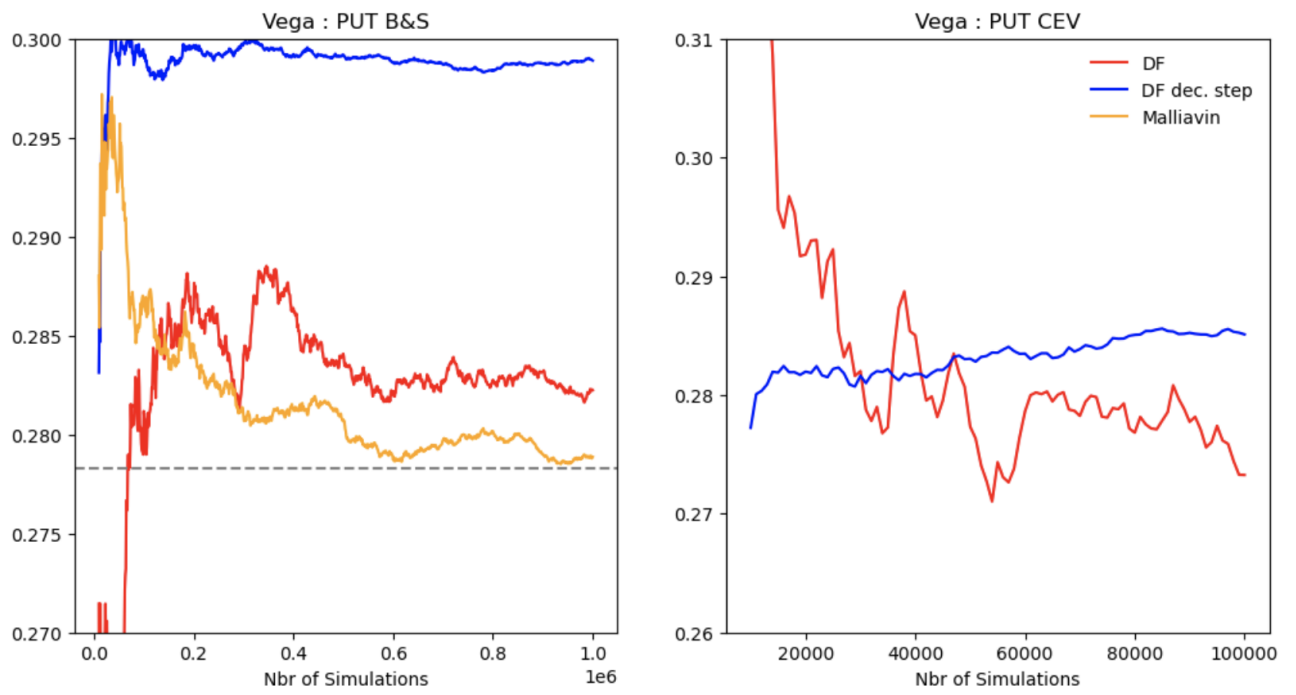


FIGURE 7 – Convergence of the different methods for Binary Put Vega

7 Conclusion

The report highlights several important findings. The finite difference method is effective only for weaker degrees of Greeks (Δ and ν) and very regular payoffs. To improve the results, a simple modification known as the finite difference method with a decreasing step can be used. The tangent flow method gives good results, but it can be implemented for only a limited number of payoffs and degrees of Greeks. The Malliavin method yields the best results in cases where there is a more irregular dynamic than the BS framework, an increase in the degree of Greek, or less regular payoffs. However, this method is challenging to implement and does not yield strong results, making it more beneficial to use the previously described methods.

An interesting approach to improve estimation accuracy is to combine different methods, such as Malliavin calculus with the finite difference method or the tangent process. This technique is known as the localization method and has been studied previously. In the case of a call option, the weight technique is not effective as the payoff is too regular and the weight is too large. To overcome this, the Malliavin method can be localized around the singularities of the payoff function to reduce the impact of the Malliavin weight in the estimation and by using the Malliavin method on the first term and the tangent process on the second one.

Thus, we observe that the first term is null when the Malliavin's weight are too big leading to a natural variance reduction. The quality of this variance reduction depends on the choice of the parameter d , which is chosen on a case-by-case basis. This method could be interesting to improve the results obtained in this study.

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